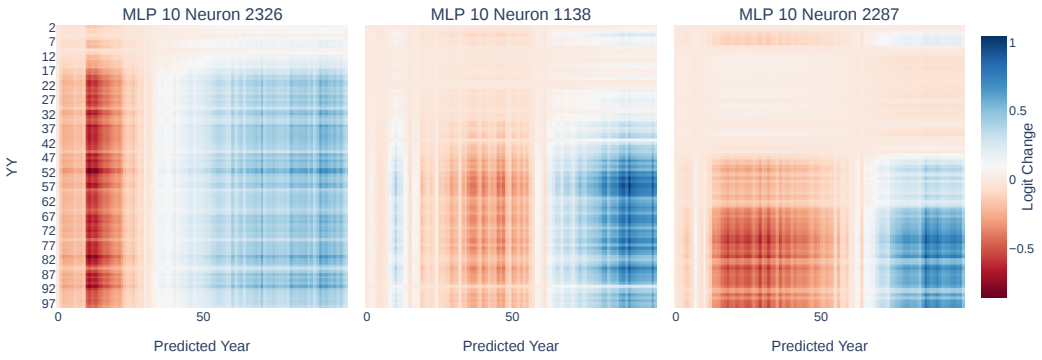

Figure 22: Direct effects of top-3 MLP 10 Neurons

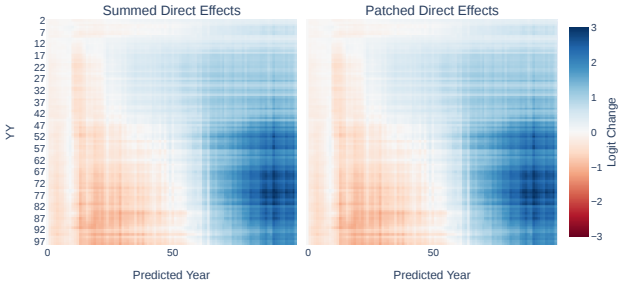

Figure 23: Summed and patched direct effects of top-10 MLP 10 Neurons

F Year-Span Circuit Generalization

In this section we provide evidence for the generalization of the year-span circuit to some (but not all) tasks. For tasks where the model failed entirely, we provide the probability heatmaps (to show its failure). For tasks where the model succeeded, or incorrectly generalized the greater-than circuit, we additionally provide path patching results and logit lens heatmaps to show how circuit structure and semantics are preserved.

Tasks Failed Figure 24 displays probability heatmaps for "1799, 1753, 1733, 1701, 16YY, 16", "1695, 1697, 1699, 1701, 1703, 17", and "17YY is smaller than 17". In the first case, GPT-2 predicts a roughly uniform distribution around YY. In the second case, the right answer varies: though the penultimate number in the provided sequence is YY="03", and the sequence increases by 2 step, depending on YY, we must vary the increase per step, in order to avoid the single-token number "1700". In any case, the model fails to predict the correct answer, often predicting YY+1, although the step is always > 1. In the final case, the model always outputs YY.

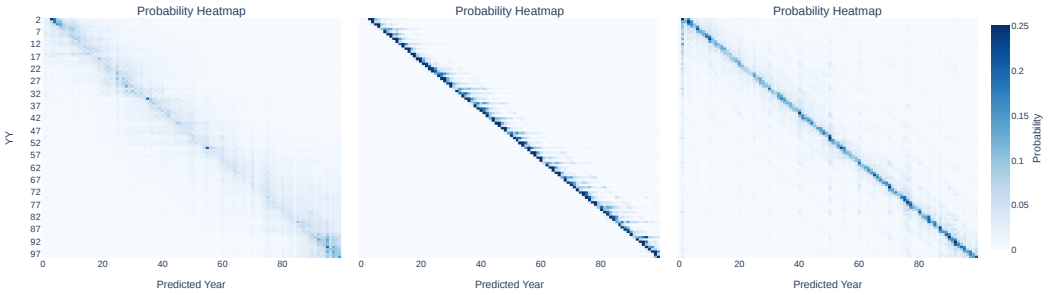

Figure 24: Probability heatmaps for (left to right) "1799, 1753, 1733, 1701, 16YY, 16", "1695, 1697, 1699, 1701, 1703, 17", and "17YY is smaller than 17".

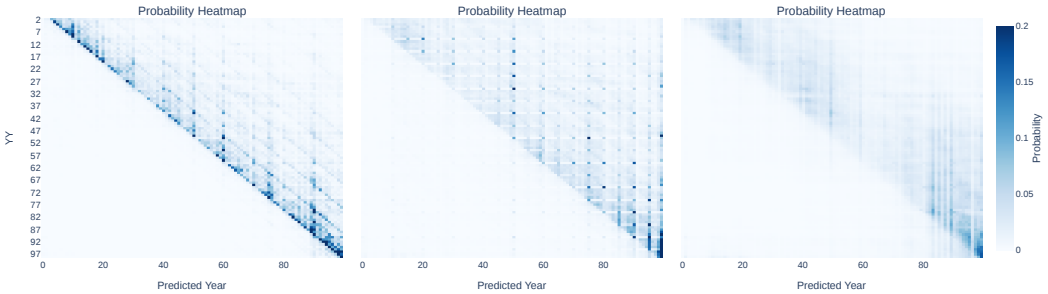

Figure 25: Probability heatmaps for (left to right) "The <noun> started in the year 17YY and ended in the year 17", "The <noun> happened in 17YY. Some years later, it is now the year 17", and "1599, 1607, 1633, 1679, 17YY, 17".

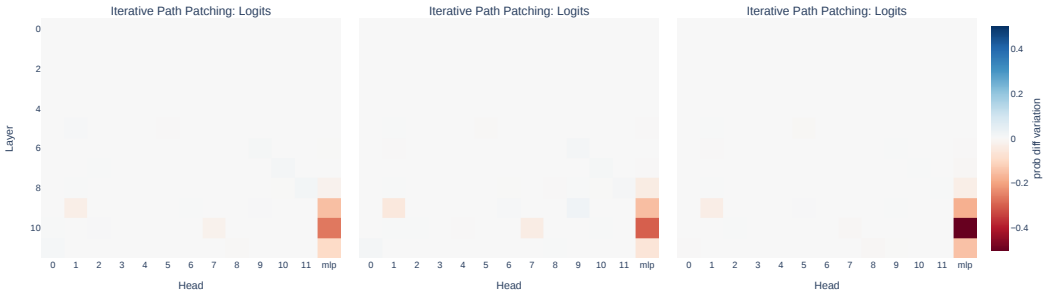

Figure 26: Iterative path patching plots (C, logits) for (left to right) "The <noun> started in the year 17YY and ended in the year 17", "The price of that <luxury good> ranges from $ 17YY to $ 17", and "1599, 1607, 1633, 1679, 17YY, 17".

Tasks Completed Correctly Figure 25 displays probability heatmaps for "The <noun> started in the year 17YY and ended in the year 17", "The price of that <luxury good> ranges from $ 17YY to $ 17", and "1599, 1607, 1633, 1679, 17YY, 17". All tasks are completed successfully, just like year-span prediction, though note that the second task is completed less well, as is visible in its heatmap. Its probability difference is only 75%, as opposed to 90%.

Given this, we proceed using iterative path patching direct logits connections as before; Figure 26 shows the results. All of these plots look almost identical to our original plots, so we evaluate the circuit on each of tasks using the methodology from Section 3.2. This works for the first two tasks, with performance recoveries $> 90\%$, but not the last.

For the last task, we observe that MLP 8 relies also on MLP 7, which in turn relies on two extra attention heads not observed in our original circuit: a7.h11 and a6.h1 (Figure 27). Accounting for this in our circuit leads us back to $> 90\%$ loss recovery.

Tasks Completed Incorrectly Finally, we address the tasks "The <noun> ended in the year 17YY and started in the year 17" and "The <noun> lasted from the year 7YY BC to the year 7", which do use our circuit, but should not do so.

Figure 28 displays probability heatmaps for "The <noun> ended in the year 17YY and started in the year 17" and "The <noun> lasted from the year 7YY BC to the year 7". Both tasks are completed successfully, though note that the latter task is completed less well.

As with the other tasks using this circuit, the iterative path patching plots (Figure 29) look similar to those of year-span prediction. Note that since the goal for these tasks is to produce "less-than", patching and impeding circuit components improves task performance (indicated in blue rather than red), since the circuit performs "greater-than". We evaluate the circuit on each of tasks using the methodology from Section 3.2. This works for both tasks, with performance recoveries $> 90\%$.

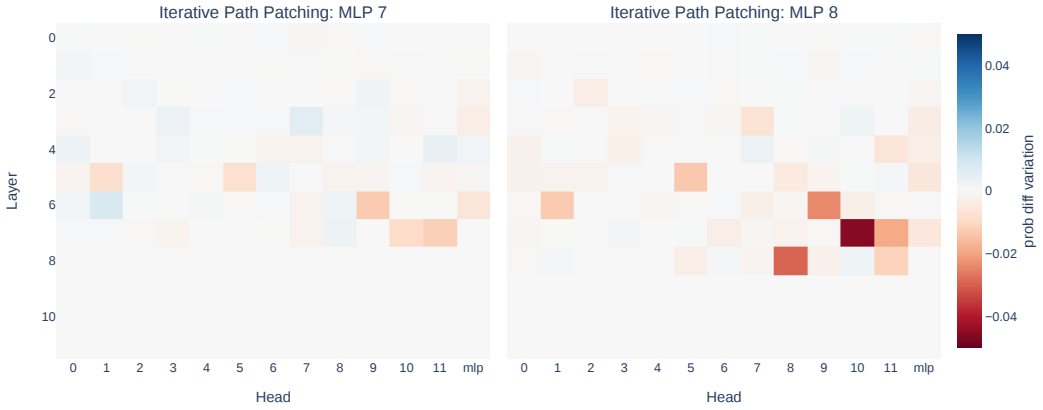

Figure 27: Iterative path patching plots for "1599, 1607, 1633, 1679, 17YY, 17", searching for components that influence the circuit via MLPs 8 (left) and 7 (right).

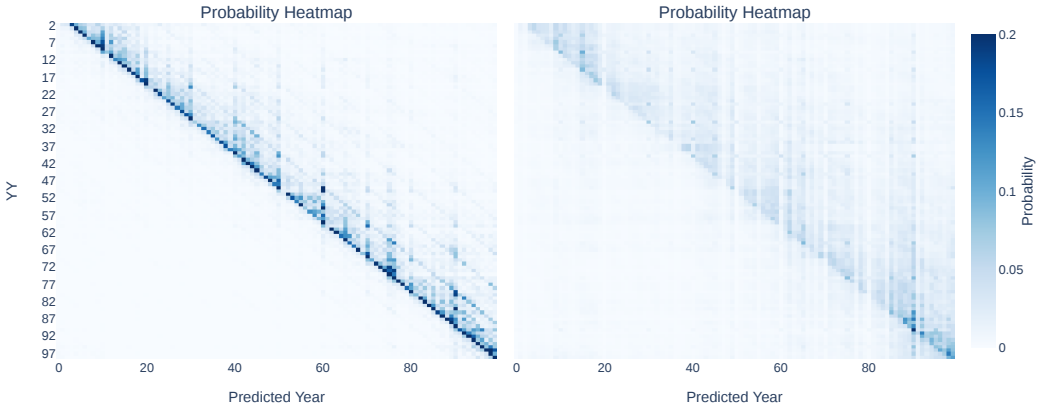

Figure 28: Probability heatmaps for "The <noun> ended in the year 17YY and started in the year 17" (left) and "The <noun> lasted from the year 7YY BC to the year 7" (right).

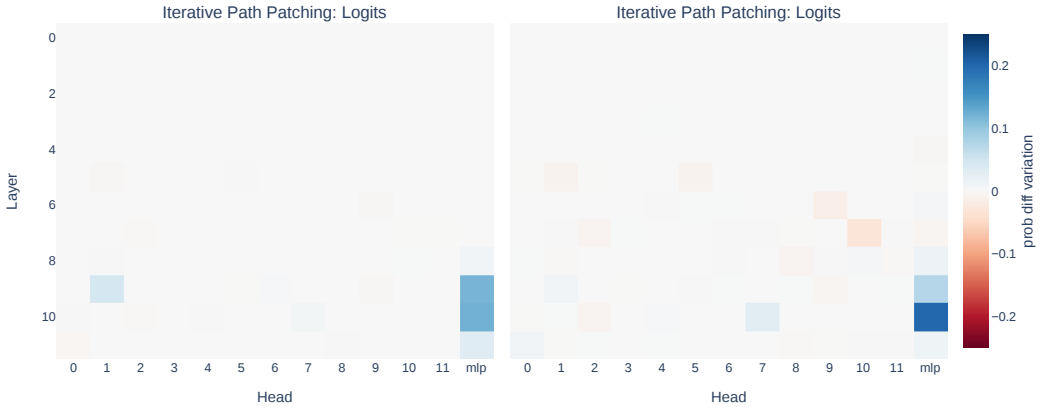

Figure 29: Iterative path patching plots (C, logits) for "The <noun> ended in the year 17YY and started in the year 17" (left) and "The <noun> lasted from the year 7YY BC to the year 7" (right).